

Program: Diploma in Fire Technology and Safety	
Course Code: 4463	Course Title: Hazard Identification and Risk Management
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week : 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To understand hazard identification techniques and risk analysis in process safety.
- To apply consequence modelling and risk estimation for hazardous scenarios.
- To study risk perception and reduction methods.
To integrate machine learning techniques in process safety and learn human reliability analysis.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2
Occupational Health & Safety		Occupational Health & Safety	3

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	CognitiveLevel
CO1	Understand the various types of hazards and apply hazard identification methods in industrial settings.	13	Understanding
CO2	Apply consequence modelling and risk estimation techniques to hazardous scenarios.	15	Applying

CO3	Understand and implement risk perception, risk reduction, and use software for risk analysis.	15	Understanding
CO4	Integrate machine learning techniques for process safety management and understand human error prediction.	15	Applying
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	1	2	2	2
CO2	3	3	3	2	3	2	2
CO3	3	3	3	3	3	3	2
CO4	3	3	3	3	3	2	2

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand the various types of hazards and apply hazard identification methods in industrial settings		
M1.01	Understand hazard types (fire, explosion, toxic gas release).	4	Understanding
M1.02	Learn hazard identification methods (PHA, FTA, ETA).	5	Applying
M1.03	Apply Fire and Explosion Hazard Ratings and analysis methods.	4	Applying

Contents:

Types of hazards: fire, explosion, toxic gas release.
Hazard identification techniques: Preliminary Hazard Analysis (PHA), Fault Tree Analysis (FTA), Event Tree Analysis (ETA).
Hazard ratings: Dow Fire and Explosion Hazard Index, Mond Index.

CO2	Apply consequence modelling and risk estimation techniques to hazardous scenarios.		
M2 No	Description	Duration (Hours)	Cognitive Level
M2.01	Learn and apply source and effect models for hazardous events.	5	Understanding
M2.02	Understand and estimate individual and societal risks.	5	Applying
M2.03	Apply software tools (ALOHA, PHAST) for effect and damage calculations.	5	Applying
	SeriesTestI	1	
Contents: Source Models: Discharge Rate Models, Flash and Evaporation Dispersion Models. Effect Models: Toxic gas effects, thermal effects, explosion effects. Risk Estimation: Individual risk, societal risk.,Software tools for consequence modelling: ALOHA, PHAST.			
CO3	Understand and implement risk perception, risk reduction, and use software for risk analysis.		
M3.01	Understand risk perception and apply ALARP for risk reduction.	5	Understanding
M3.02	Learn risk reduction measures and their presentation.	5	Applying
M3.03	Use software tools for risk analysis and case studies.	5	Applying
Contents: Risk perception, risk targets, tolerability, and acceptability. ALARP concept (As Low As Reasonably Practicable). Risk contour presentation, Bow-Tie Analysis, Dynamic Risk Analysis.			
CO4	Integrate machine learning techniques for process safety management and understand human error prediction		
M4.01	Understand machine learning fundamentals and its applications in process safety.	5	Understanding
M4.02	Learn about machine learning for fault detection and diagnosis.	5	Applying

M4.03	Understand Human Reliability Analysis (HRA) and techniques for human error prediction.	5	Understanding
	SeriesTest II	1	
Contents: Machine Learning Fundamentals: Empirical Risk Minimization, Maximum Likelihood Estimation. Machine Learning Algorithms for Fault Detection, Process Safety.Human Reliability Analysis (HRA) and techniques for error prediction: THERP (Technique for Human Error Rate Prediction) Software applications: ALOHA, PHAST.			

Text/Reference:

T/R	BookTitle/Author
T1	Guidelines for Hazard Evaluation Procedures by AIChE/CCPS (3rd Edition)
T2	Machine Learning in Chemical Safety and Health: Fundamentals with Applications by Qingsheng Wang and ChangjieCai
R1	Dynamic Risk Analysis in Chemical and Petroleum Industries by Nicola Paltrinieri and Faisal Khan
R2	Lee's Loss Prevention in the Process Industries (4th Edition) edited by Sam Mannan
R3	Major Hazard Control: A Practical Manual by ILO (1993)